

News Trend Analysis using Apache Spark and Kafka

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INTRODUCTION

The volume of news data generated daily is extremely large, making it difficult to analyze manually using traditional methods. With the rapid growth of digital media, there is a need for efficient tools that can handle and process such large datasets. Big Data technologies like Apache Spark and Kafka enable fast and scalable data processing. This project focuses on analyzing news trends by processing large-scale data and extracting meaningful insights such as patterns over time, source activity, and category distribution.

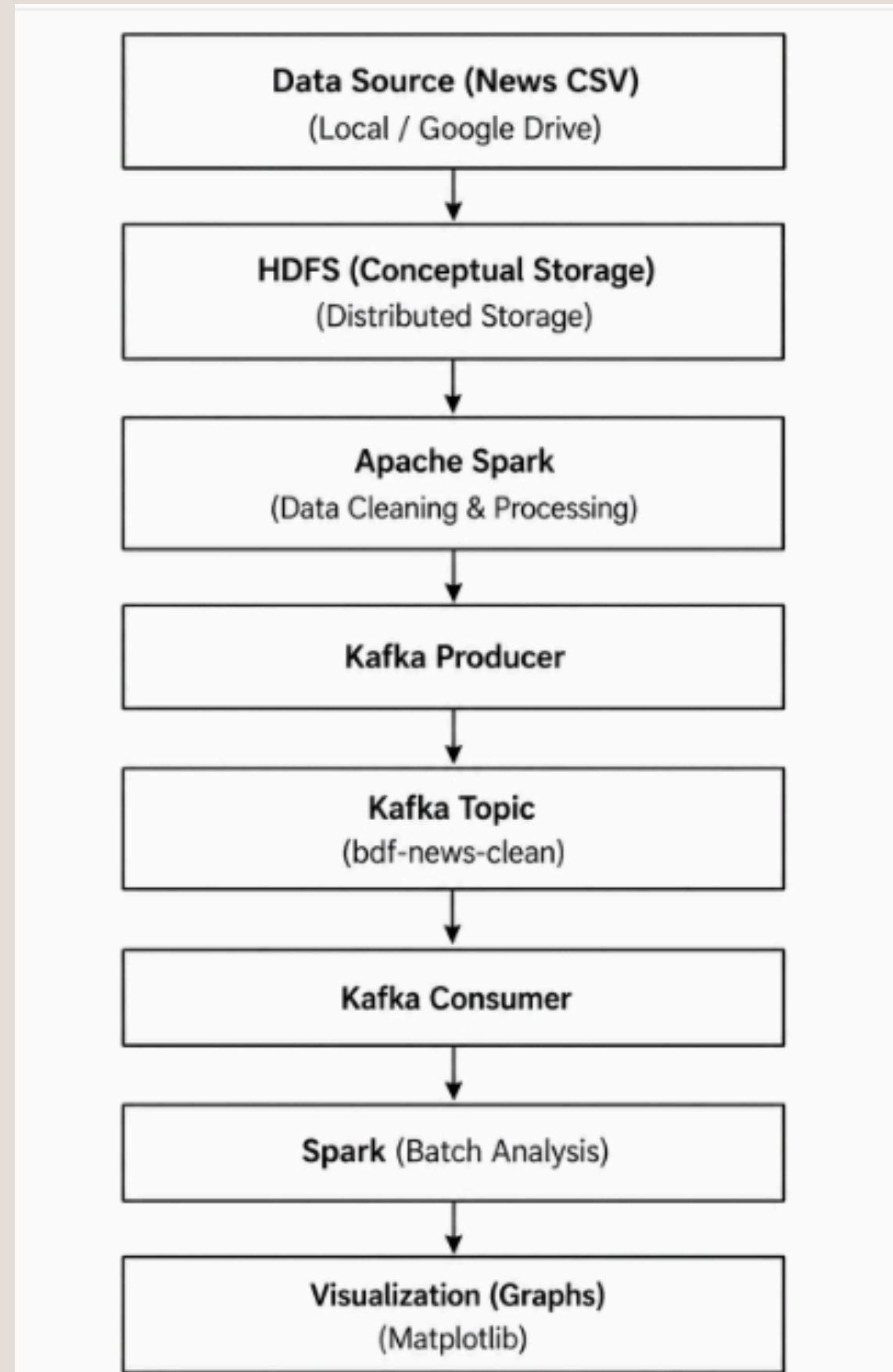
PROBLEM STATEMENT

Traditional data processing systems are not capable of efficiently handling the large-scale volume of news data generated every day. As the size and complexity of data increase, there is a strong need for scalable solutions that provide efficient processing and storage. Additionally, extracting meaningful insights such as trends and patterns from news articles becomes challenging without proper tools. This project aims to address these challenges by analyzing news data to identify trends over time and understand the publishing behavior of different news sources.

OBJECTIVES

- Process large-scale news data efficiently using Apache Spark
- Perform data cleaning and preprocessing to ensure accuracy
- Analyze trends over time (year-wise and month-wise)
- Identify and visualize patterns in news sources and categories

SYSTEM ARCHITECTURE



DATA PREPROCESSING

- Loaded dataset with proper parsing (handled multiline text and quotes)
- Selected relevant columns (title, publication, section, date)
- Removed unnecessary columns and noise
- Handled missing values:
 - Dropped rows with missing title and publication
 - Replaced missing section values with Unknown
- Filtered valid date records using pattern matching
- Extracted year and month from date column
- Removed original date column after extraction
- Ensured consistent data types (year and month as integers)
- Eliminated incorrect rows caused by CSV issues
- Cached cleaned dataset for efficient processing

DATA PIPELINE

- Raw news dataset loaded and cleaned using Apache Spark
- Cleaned data transformed into structured format
- Data streamed to Kafka using a Producer
- Messages stored in Kafka topic (bdf-news-clean)
- Kafka Consumer reads data from the topic
- Data processed in batches for efficient handling
- Batch data converted into Spark DataFrame
- Analytical processing performed on structured data
- Results used for trend analysis and visualization

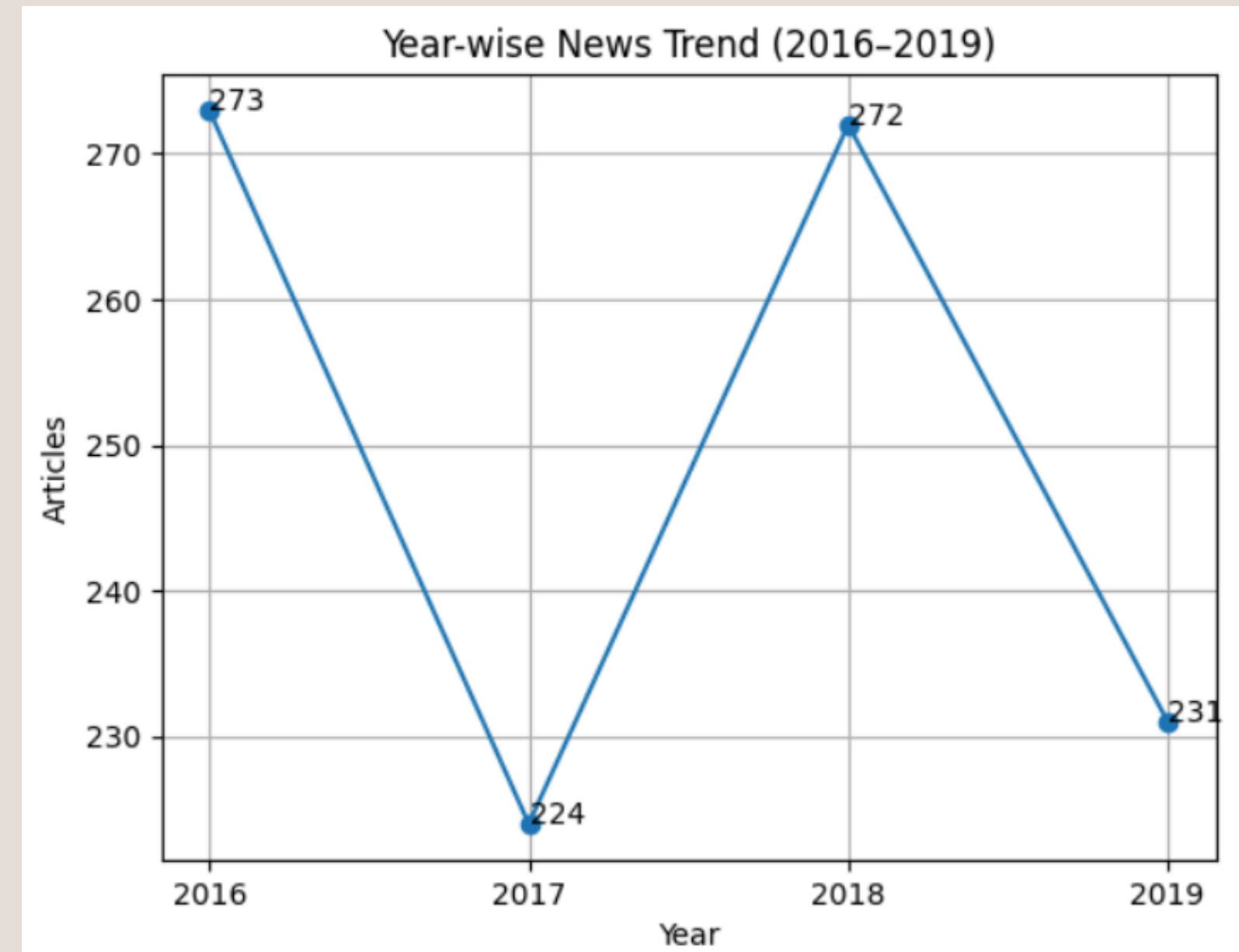
CSV → Spark → Kafka Producer → Kafka Topic → Kafka Consumer → Spark → Analysis → Visualization

ANALYSIS

- Year-wise trend analysis to identify changes over time
- Month-wise trend analysis to observe seasonal patterns
- Source-wise analysis to identify most active news publishers
- Category-wise distribution to understand content spread across sections

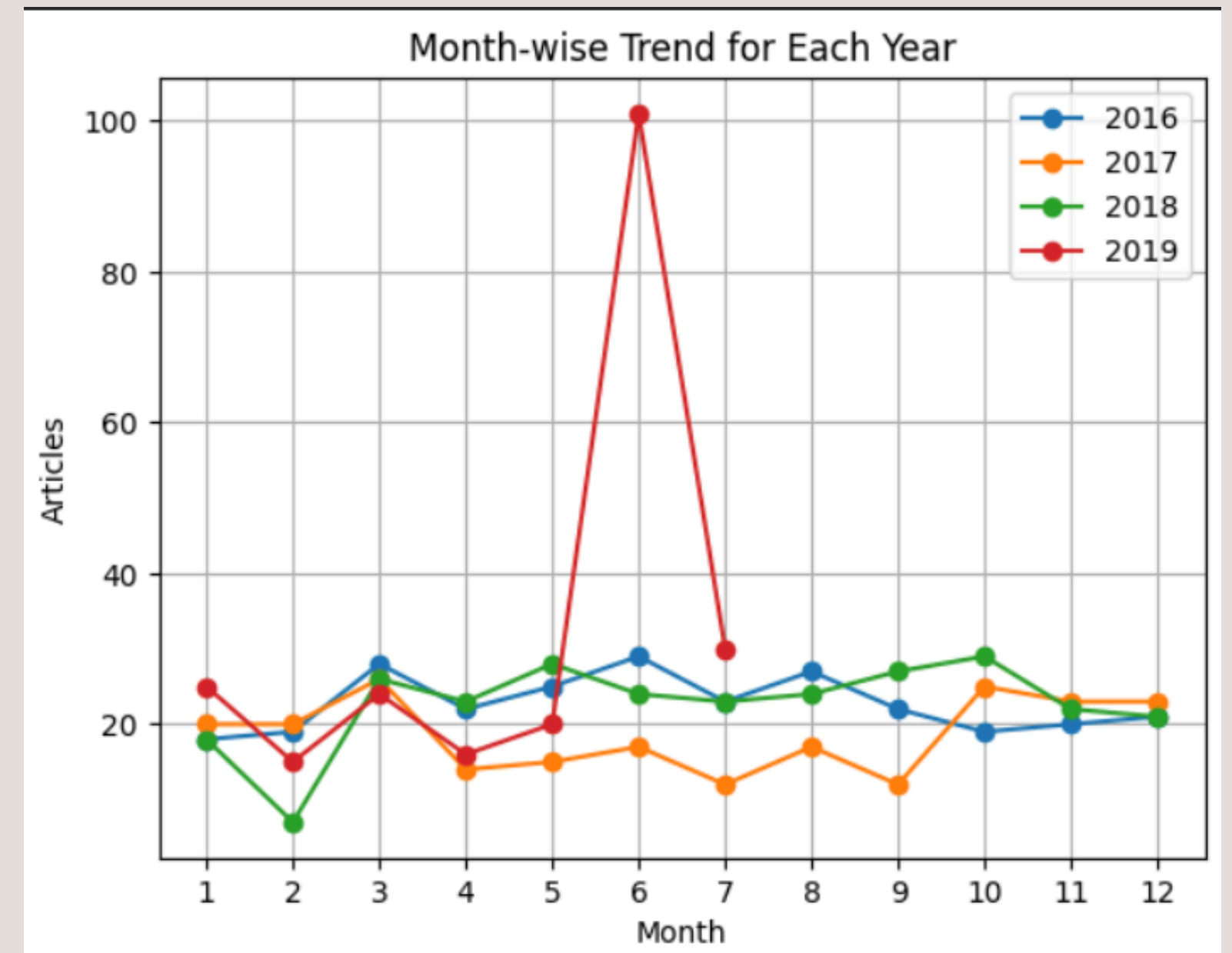
YEAR-WISE TREND

- 2016 recorded a high number of articles (~273)
- Significant drop observed in 2017 (~224)
- Peak activity again in 2018 (~272)
- Decline observed in 2019 (~231)
- Overall trend indicates fluctuations rather than steady growth



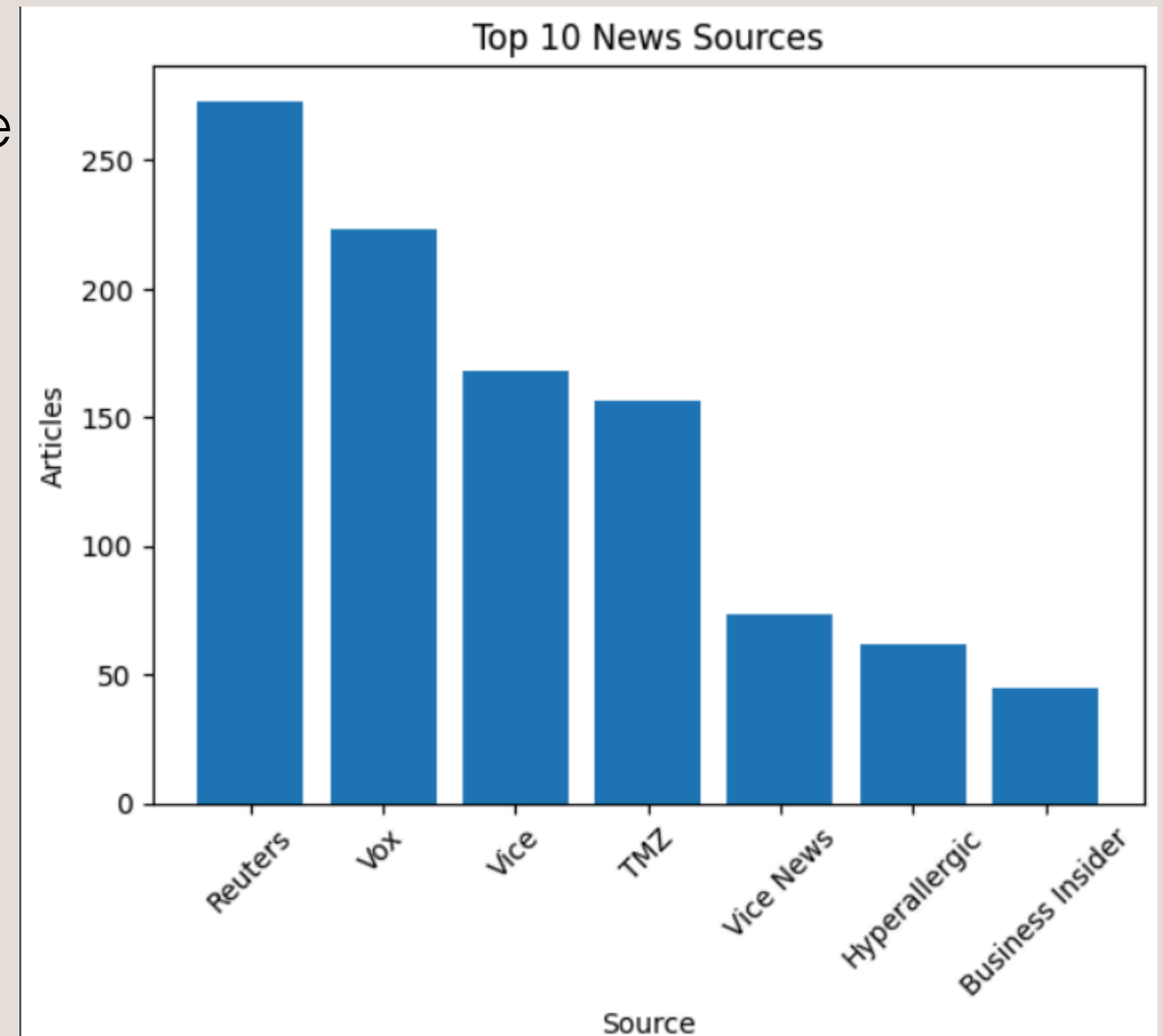
MONTH-WISE TREND

- Most months show a stable range (15–30 articles)
- 2019 shows a sharp peak in June (~100 articles)
- February 2018 shows a significant drop compared to other months
- 2016 and 2018 show relatively consistent trends
- 2017 has comparatively lower and stable activity
- Overall pattern indicates seasonal fluctuations in news publishing



SOURCE ANALYSIS

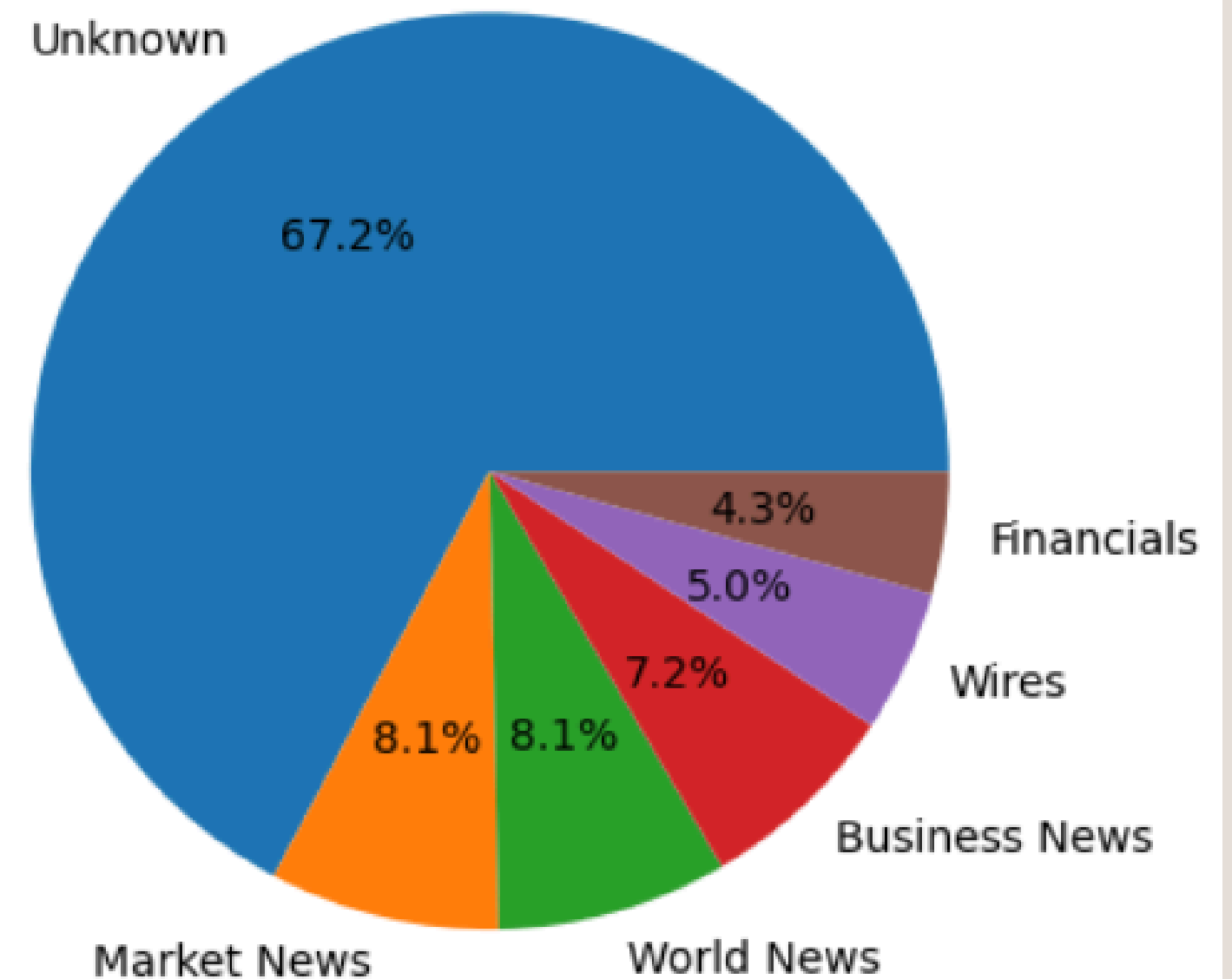
- Reuters is the most active news source with the highest number of articles
- Vox is the second most contributing publisher
- Vice and TMZ show moderate publishing activity
- Other sources like Vice News, Hyperallergic, and Business Insider contribute fewer articles



CATEGORY ANALYSIS

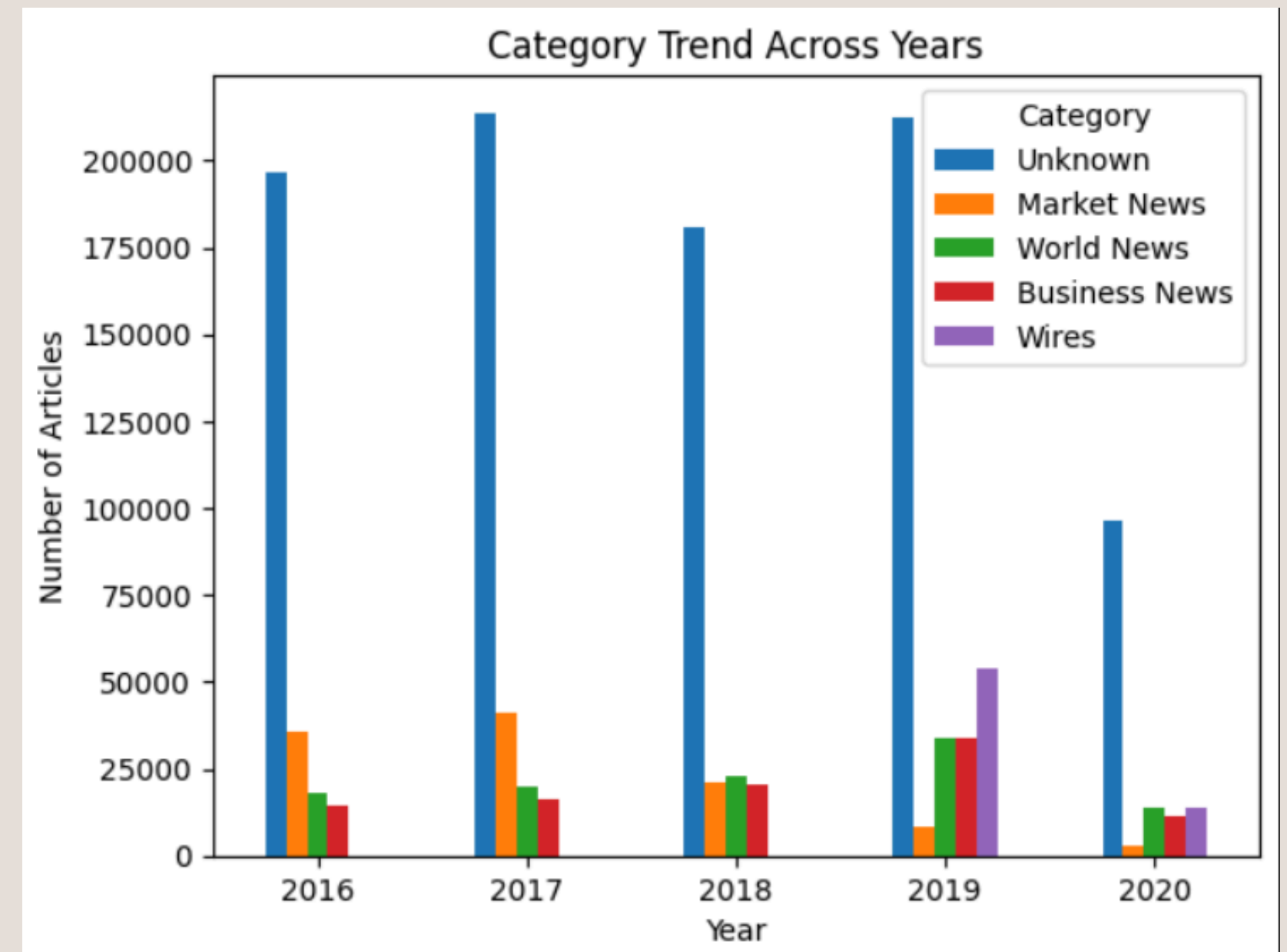
- Unknown category dominates (~67%), indicating many missing or undefined sections
- Among known categories, Market News and World News have the highest share (~8% each)
- Business News also contributes a significant portion (~7%)
- Categories like Wires and Financials have smaller contributions
- The distribution shows that a large portion of data lacks proper categorization

Top Categories Distribution



CATEGORY TREND ACROSS YEARS

- Unknown category dominates across all years, indicating large missing section data
- Market News shows high activity in 2016–2017, but declines in later years
- World News shows steady growth, especially noticeable in 2019
- Business News increases gradually, with higher values in later years
- Wires category shows a sharp spike in 2019, indicating sudden increase in that year



LIMITATIONS

- Dataset is limited to a few years (2016–2019)
- Presence of missing values (handled as “Unknown”) affects category analysis
- Kafka processing used limited batch size, not full real-time streaming
- Hadoop (HDFS) not implemented for distributed storage
- Dataset contains inconsistent or noisy entries due to CSV formatting issues

FUTURE WORK

- Integrate real-time live news data for continuous analysis
- Apply machine learning techniques for trend prediction and forecasting
- Improve data quality handling by reducing “Unknown” category values
- Optimize performance for handling larger datasets efficiently
- Extend analysis to include user behavior or engagement metrics

CONCLUSION

This project successfully demonstrates the use of big data technologies for processing and analyzing large-scale news datasets. Apache Spark was used to efficiently clean and preprocess the data, while Kafka was integrated to simulate a streaming data pipeline. Through this approach, meaningful insights were extracted by performing year-wise and month-wise trend analysis, as well as examining the contribution of different news sources and categories. The use of visualizations helped in clearly understanding patterns and variations in the data. Overall, the project highlights the importance of scalable tools in handling large datasets and shows how they can be effectively used to generate valuable insights from real-world data.



THANK YOU